

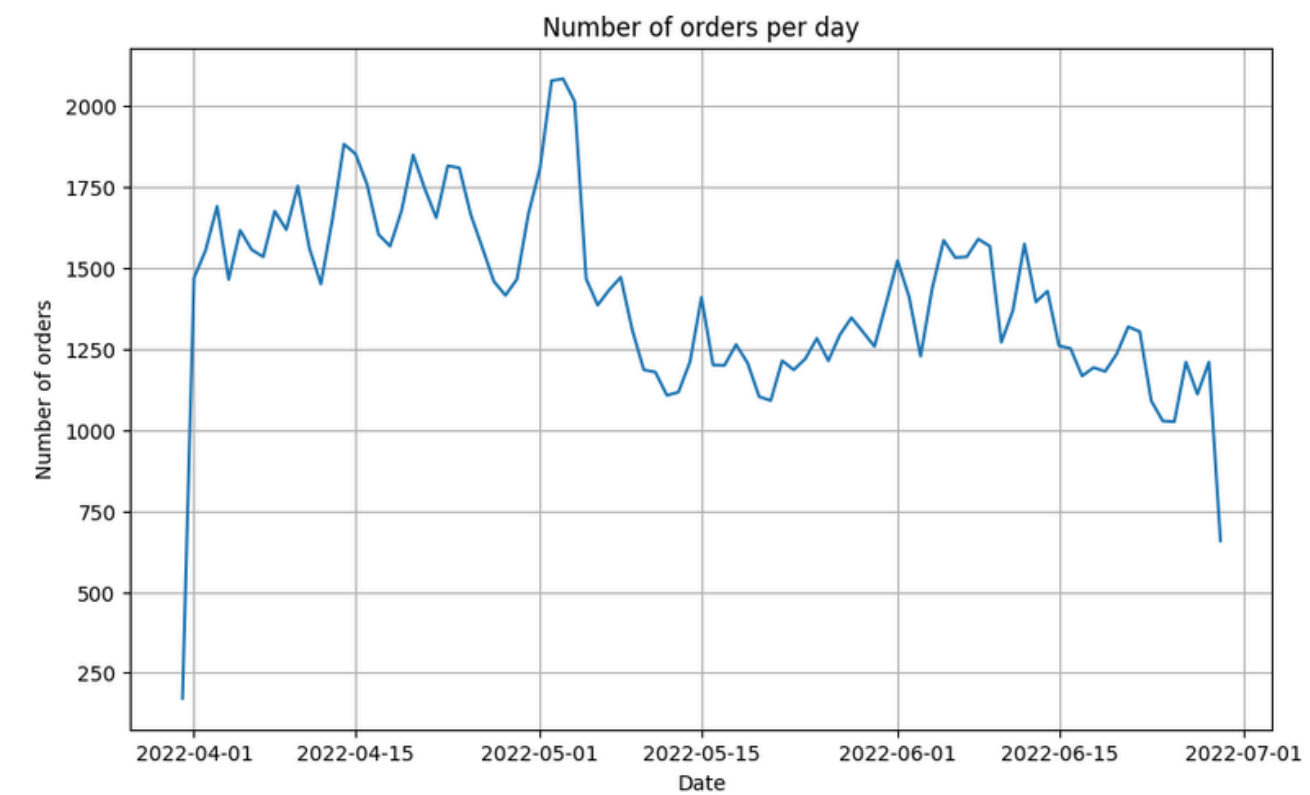
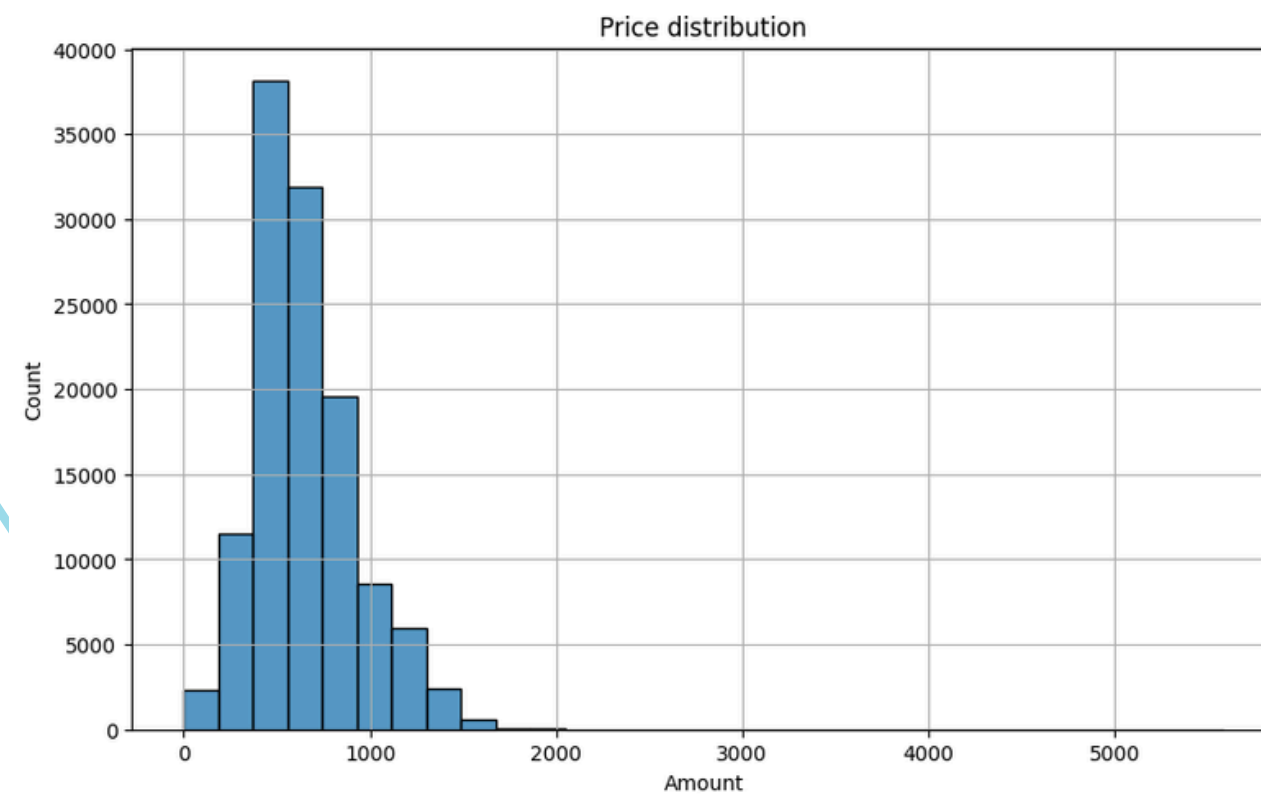
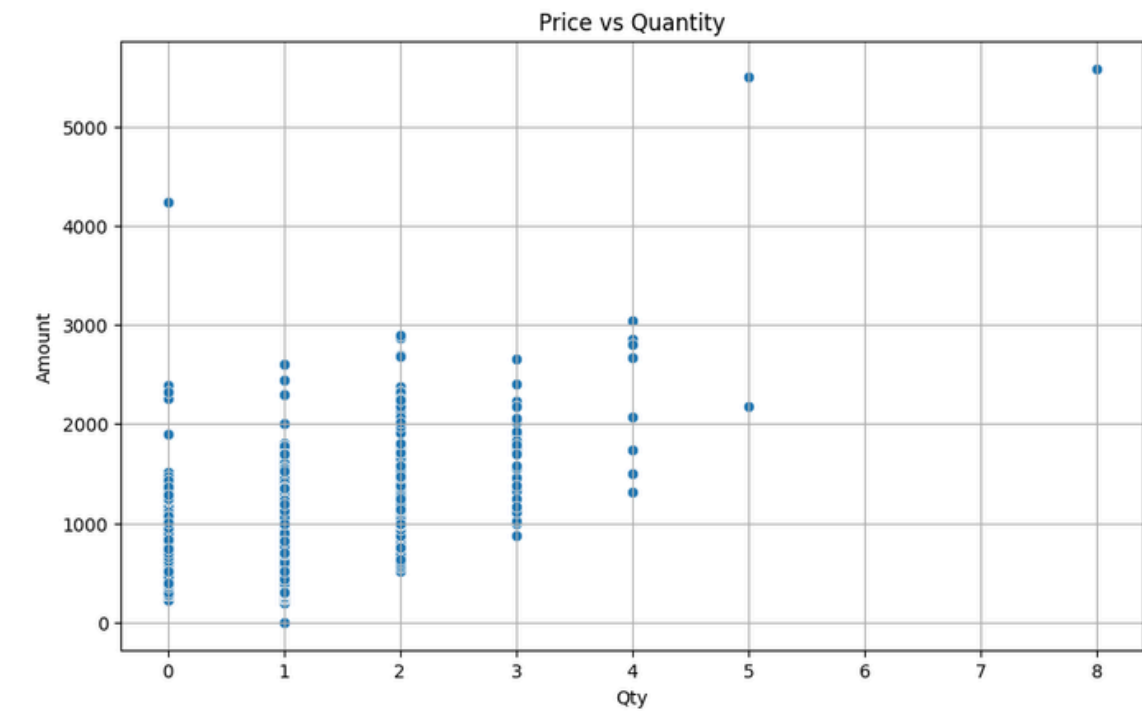
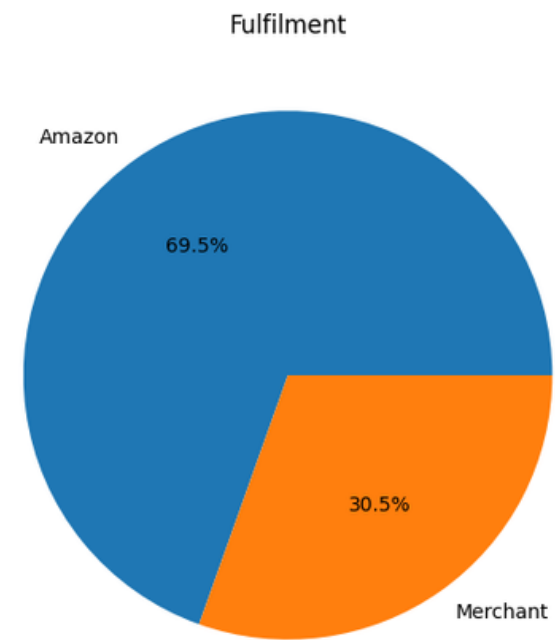
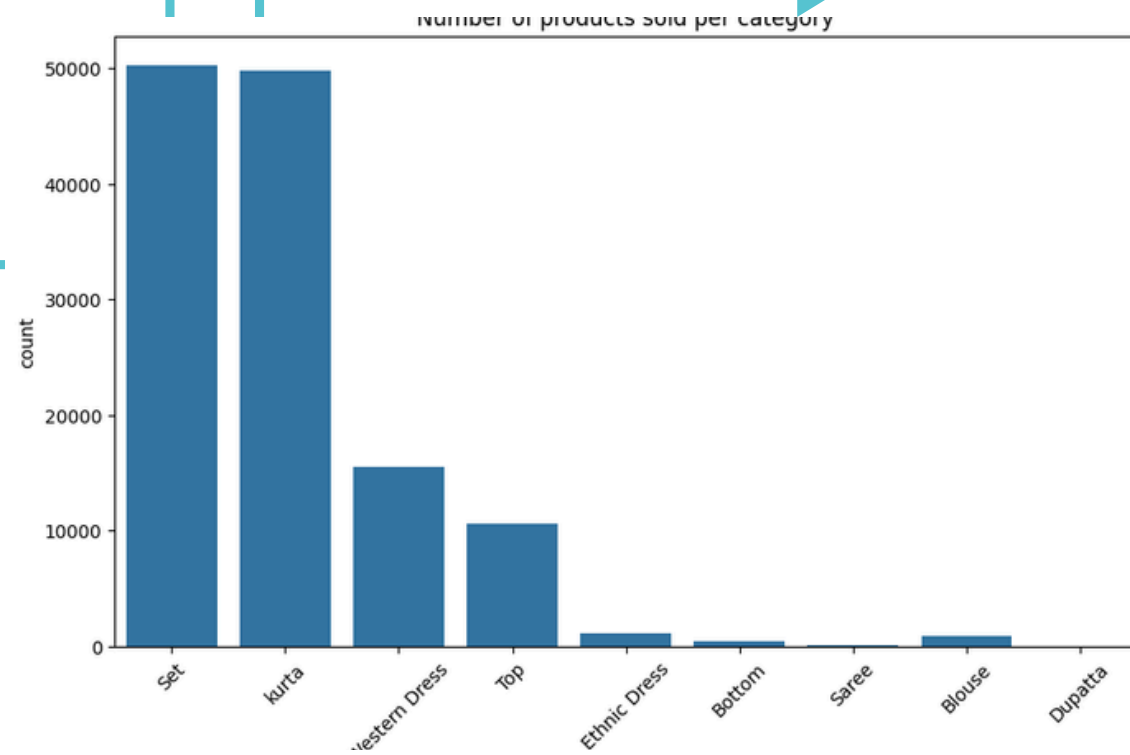
Prof. Pranav Nerurkar

ADVANCED DATA VISUALIZATION

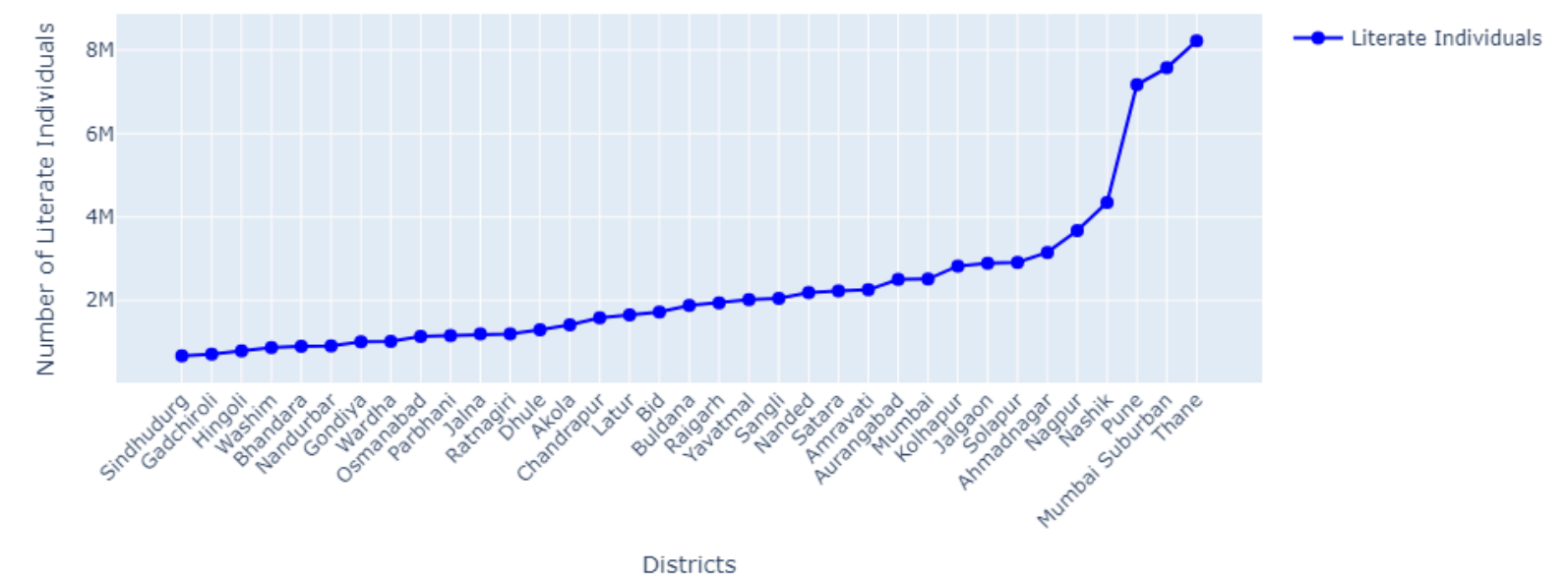
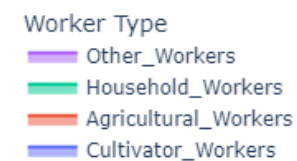
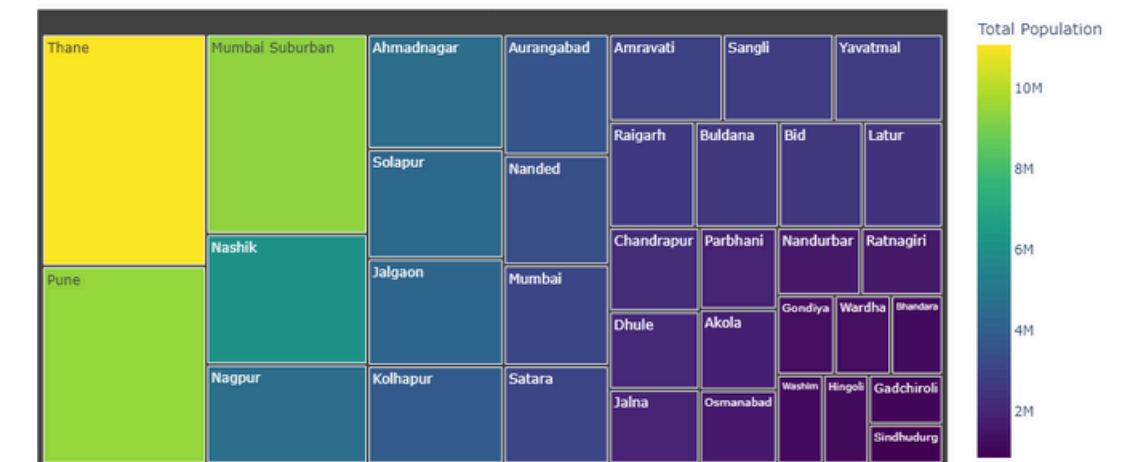
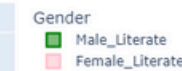
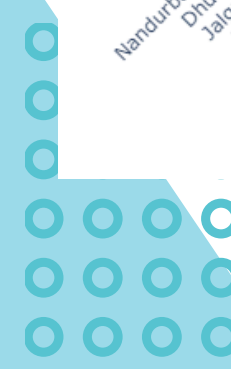
Lab Experiments

Vineet Parmar
2021300092
BE COMPS B

EXPERIMENT 1: PYTHON



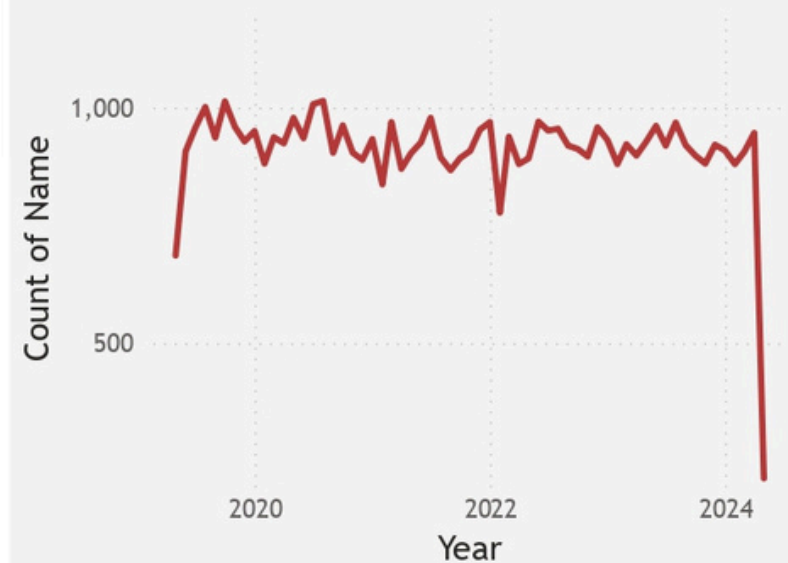
A decorative graphic consisting of a 6x6 grid of teal circles. A diagonal line runs from the top-left to the bottom-right, separating the grid into two triangular sections. The circles in the top-left section are white with teal outlines, while the circles in the bottom-right section are solid teal.



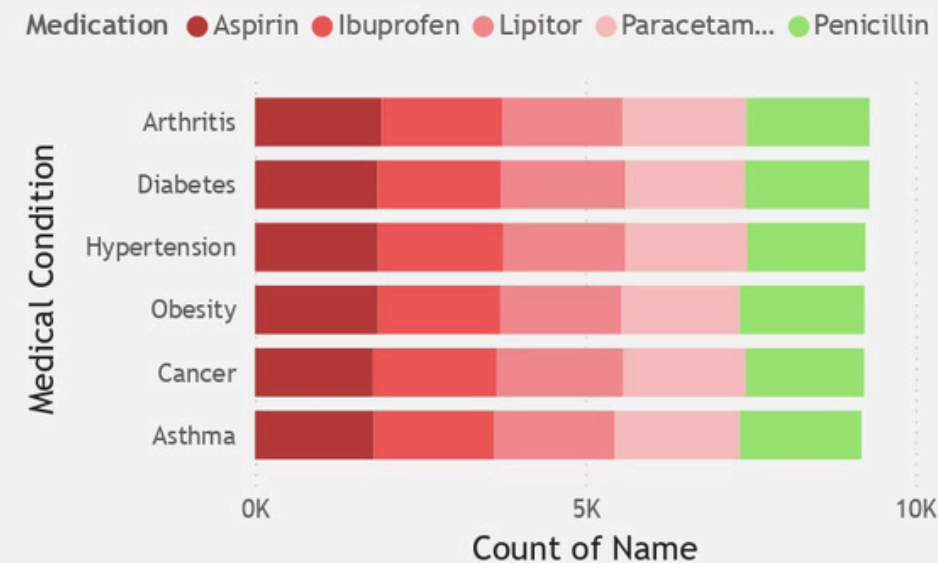
EXPERIMENT 3: POWER-BI

Executive Summary - Healthcare Report

Number of Admissions



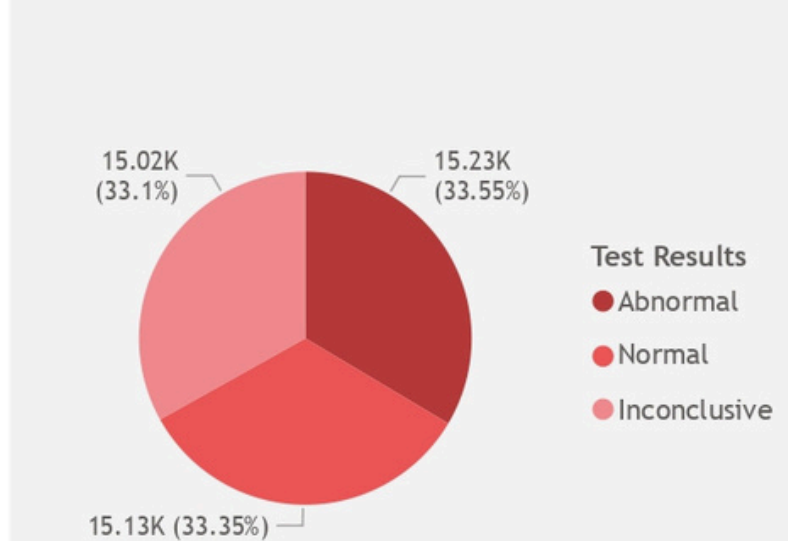
Usage of Medications



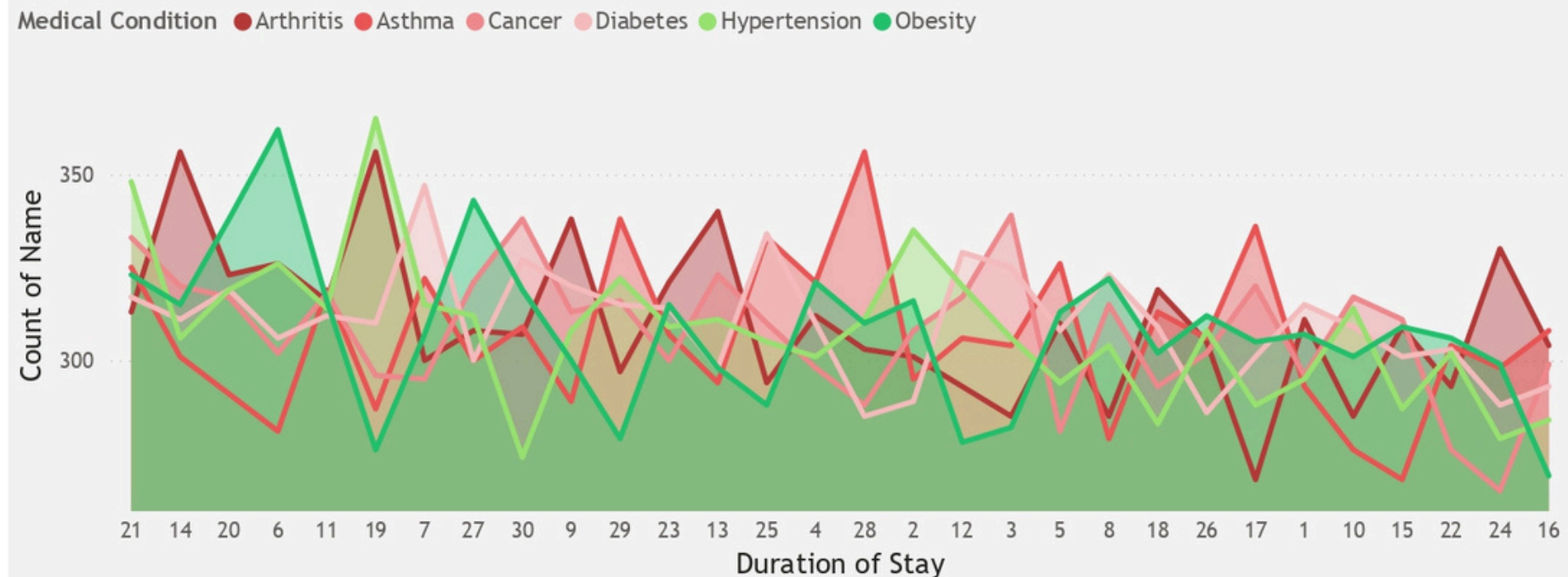
Distribution of Blood Type By Gender



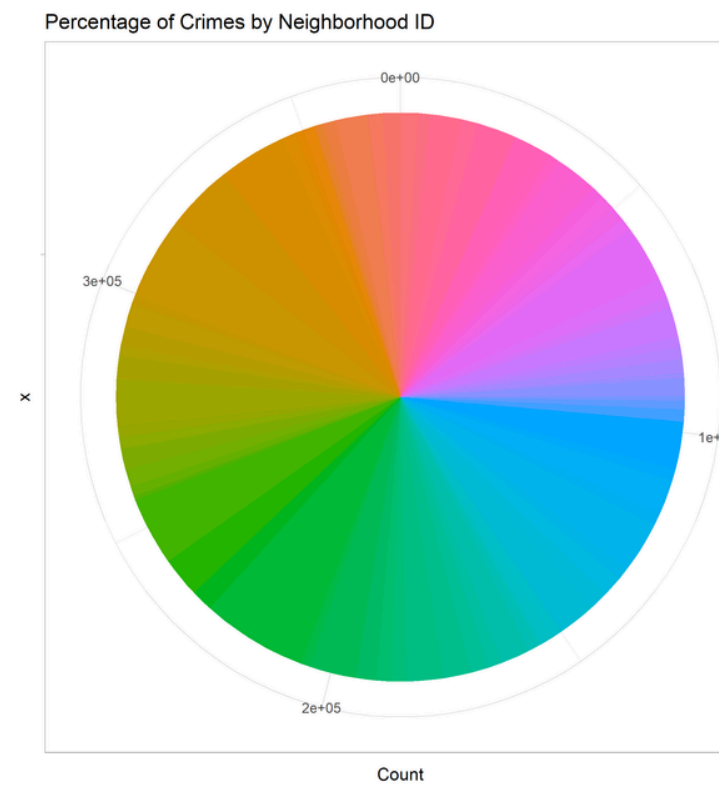
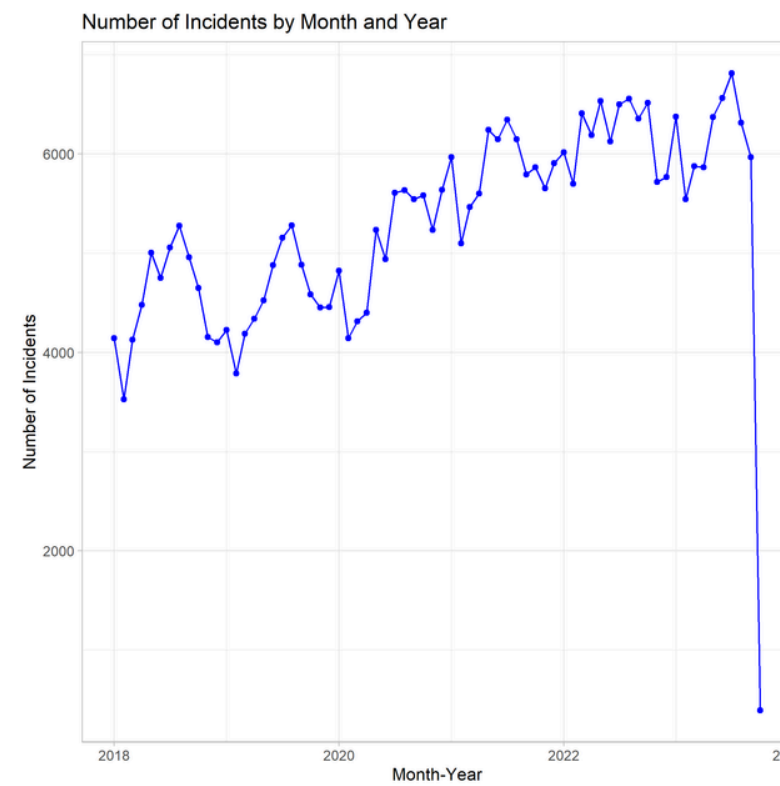
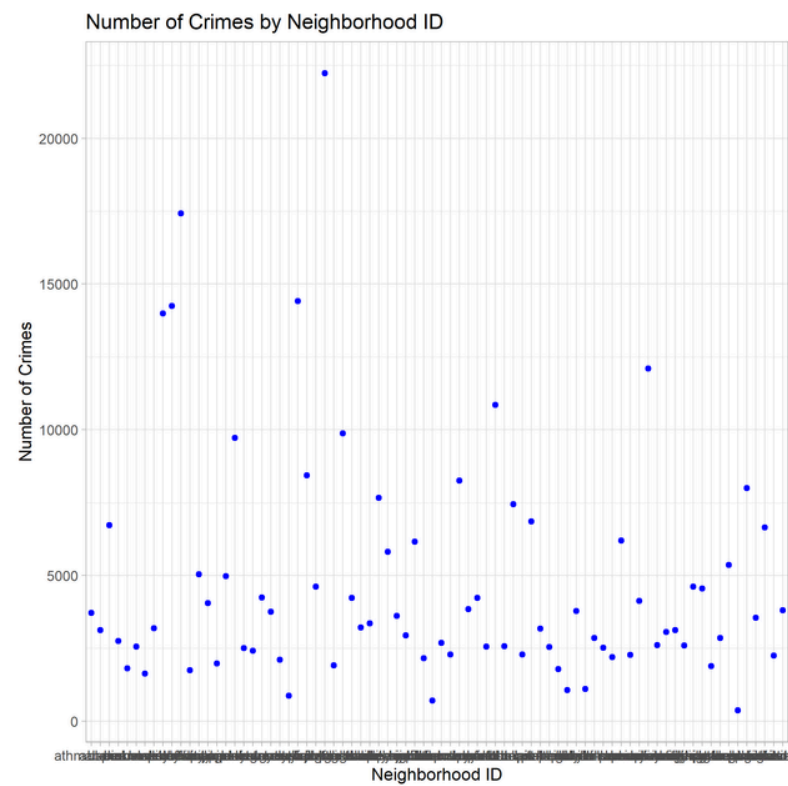
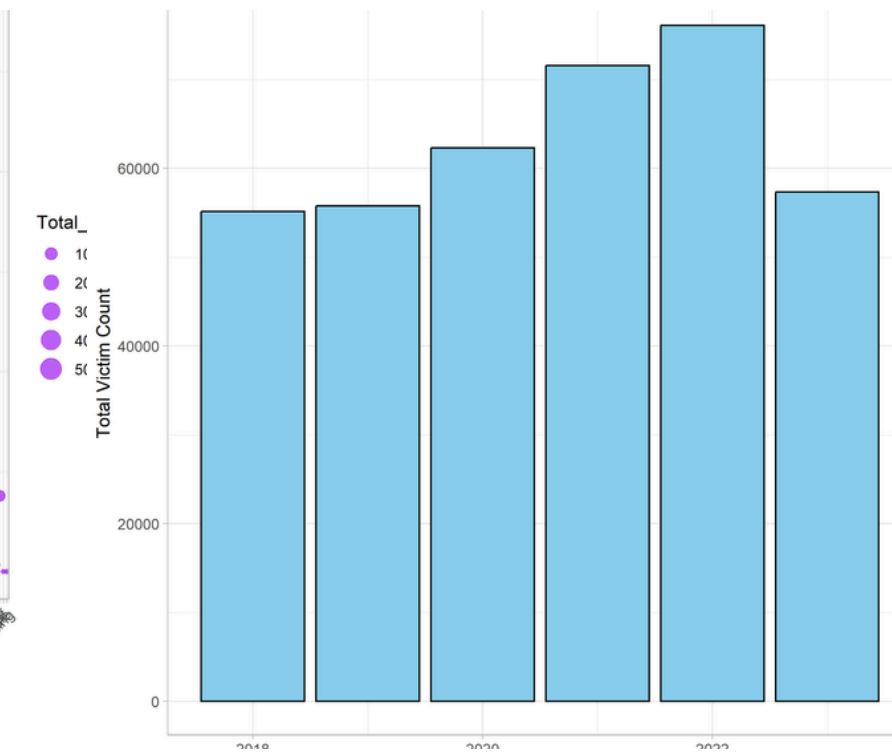
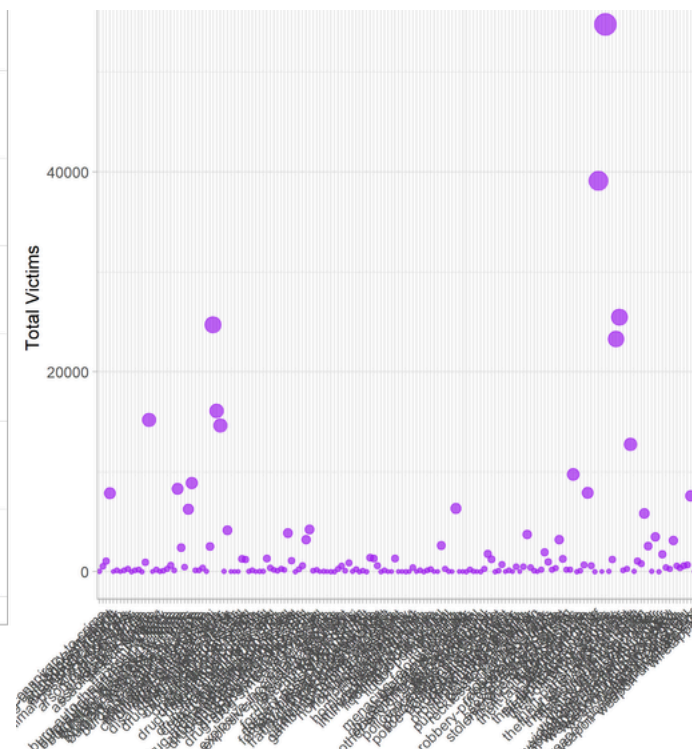
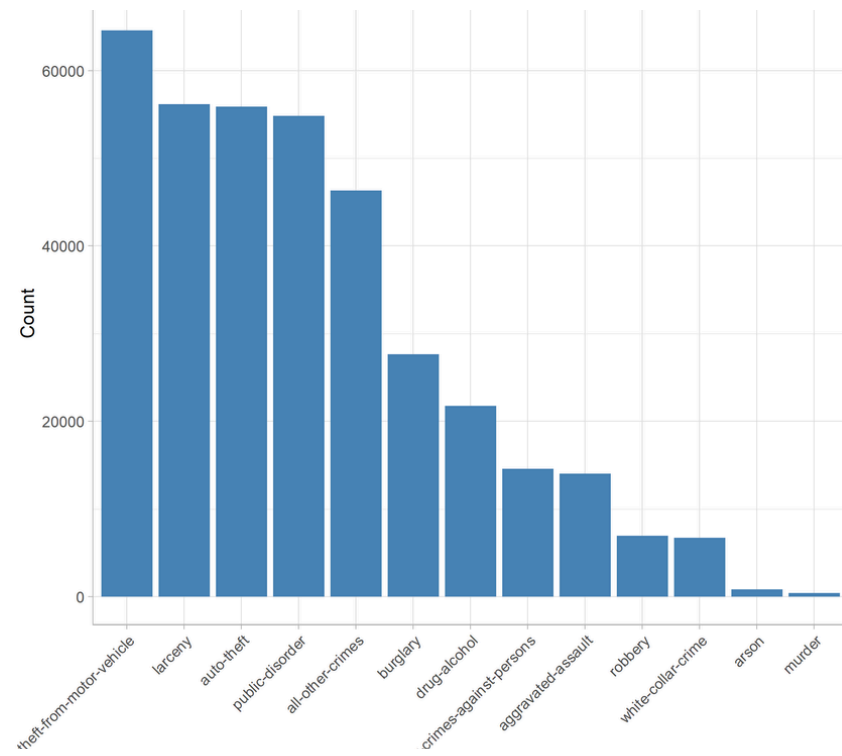
Distribution Of Test Results



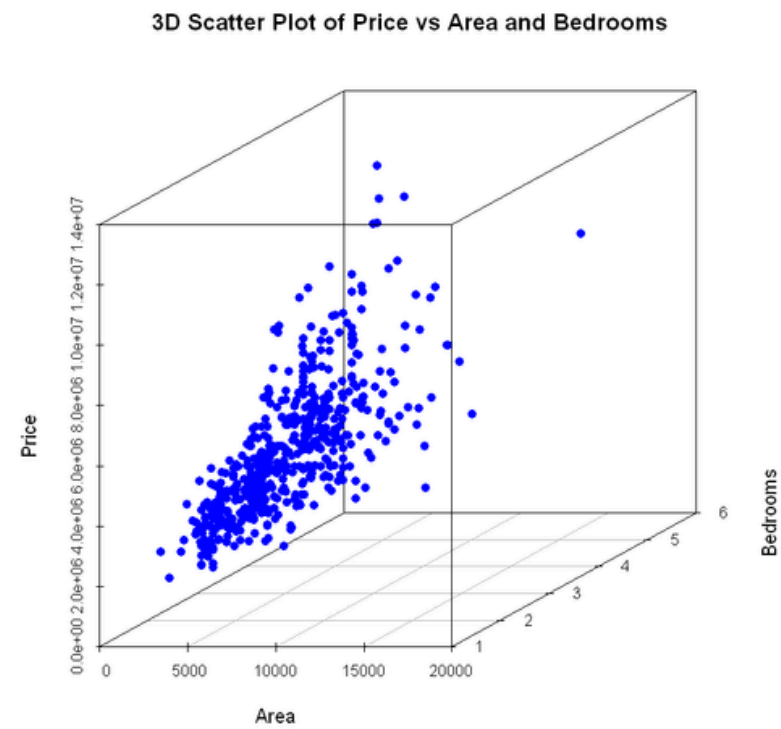
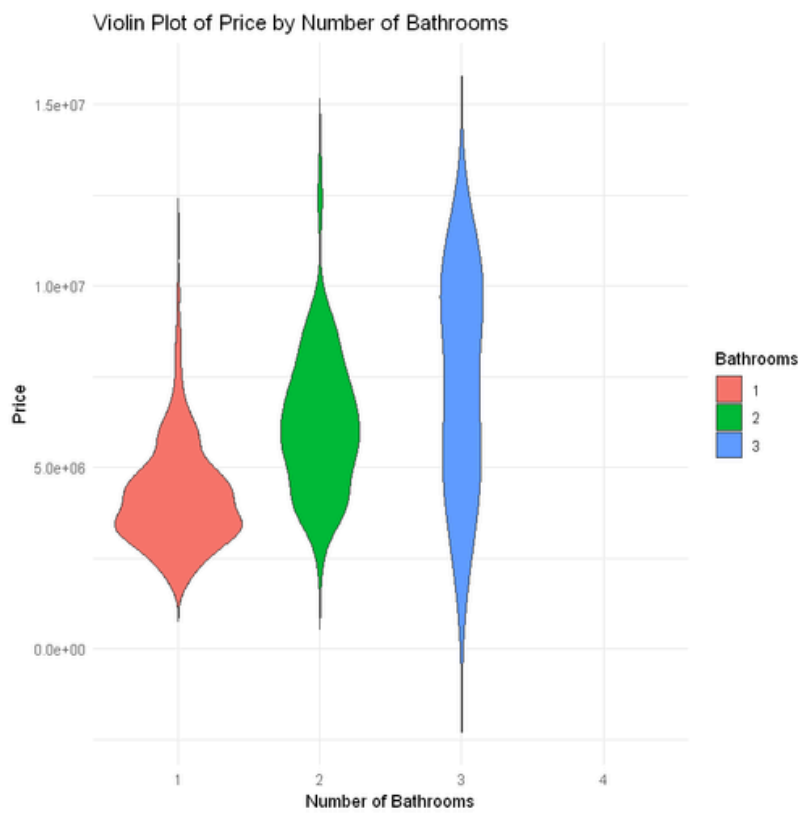
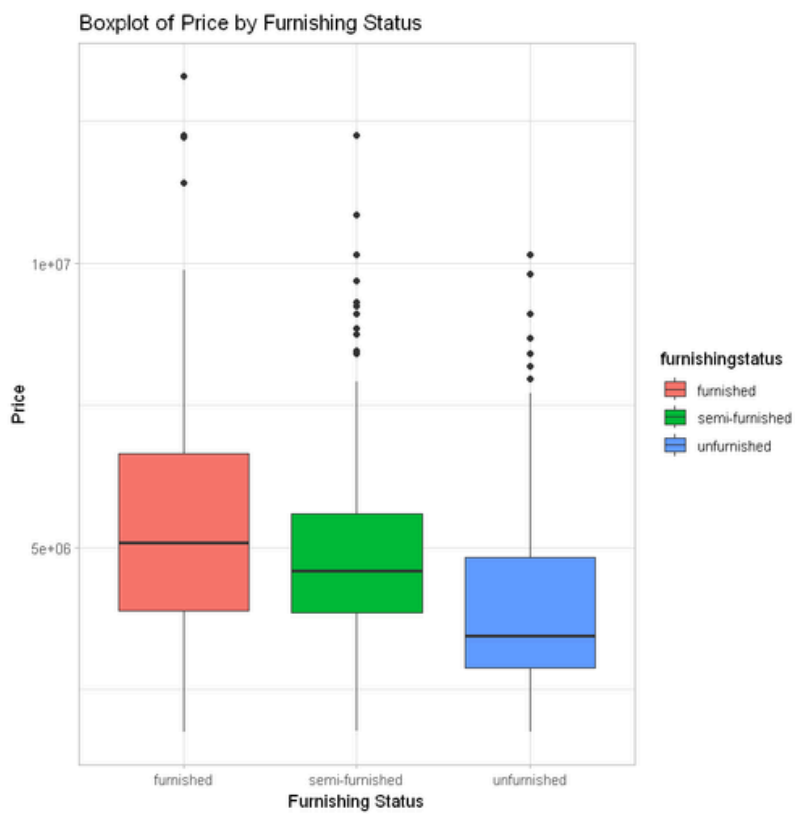
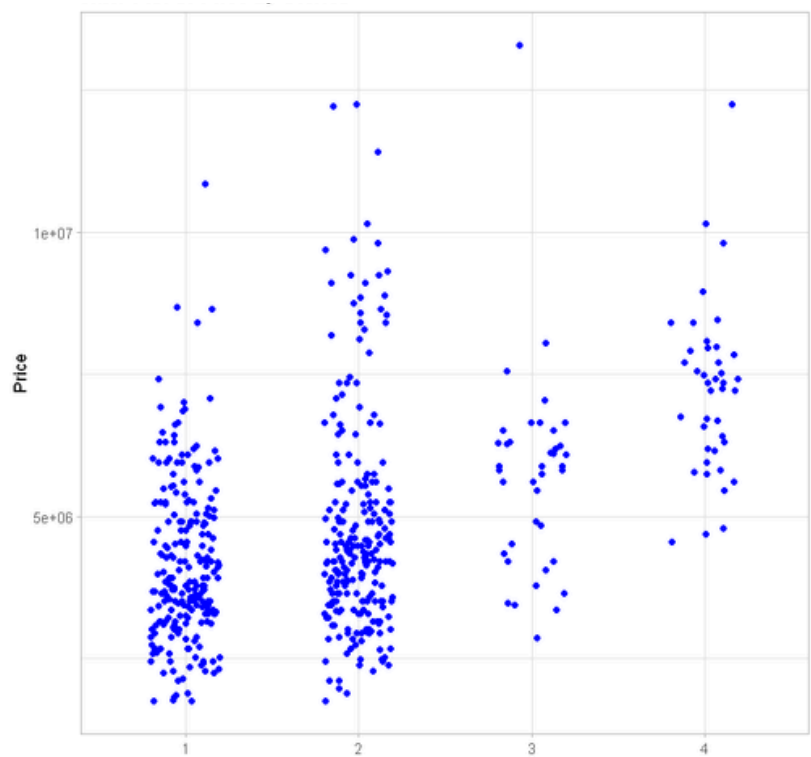
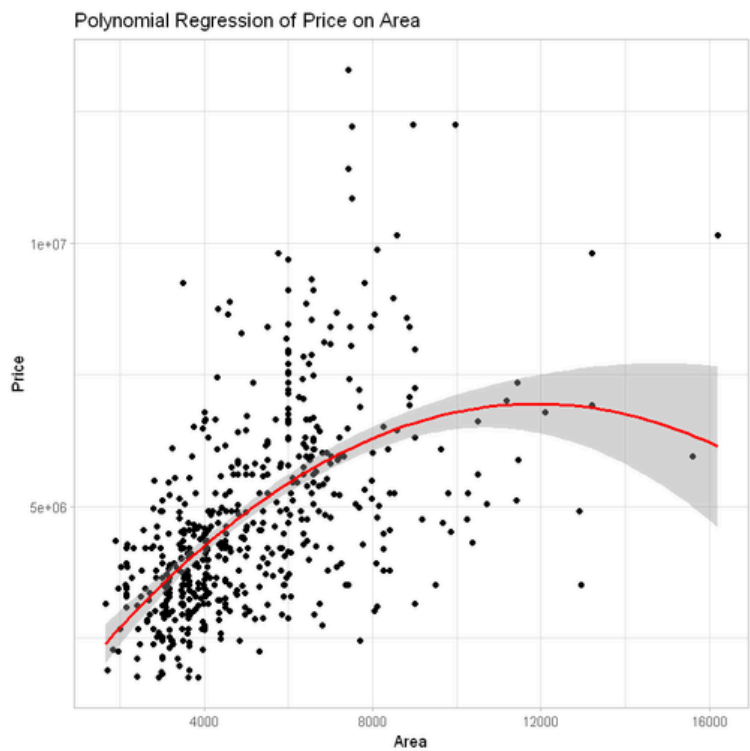
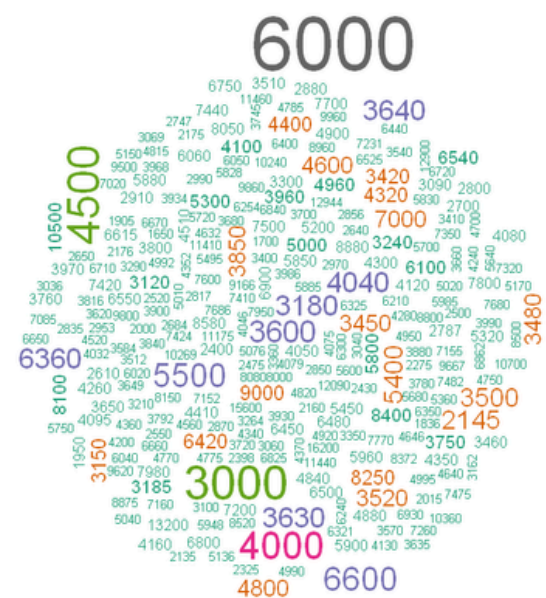
Duration of Stay According to Condition



EXPERIMENT 4: R BASIC



EXPERIMENT 5: R ADV



EXPERIMENT 6: POWER-BI

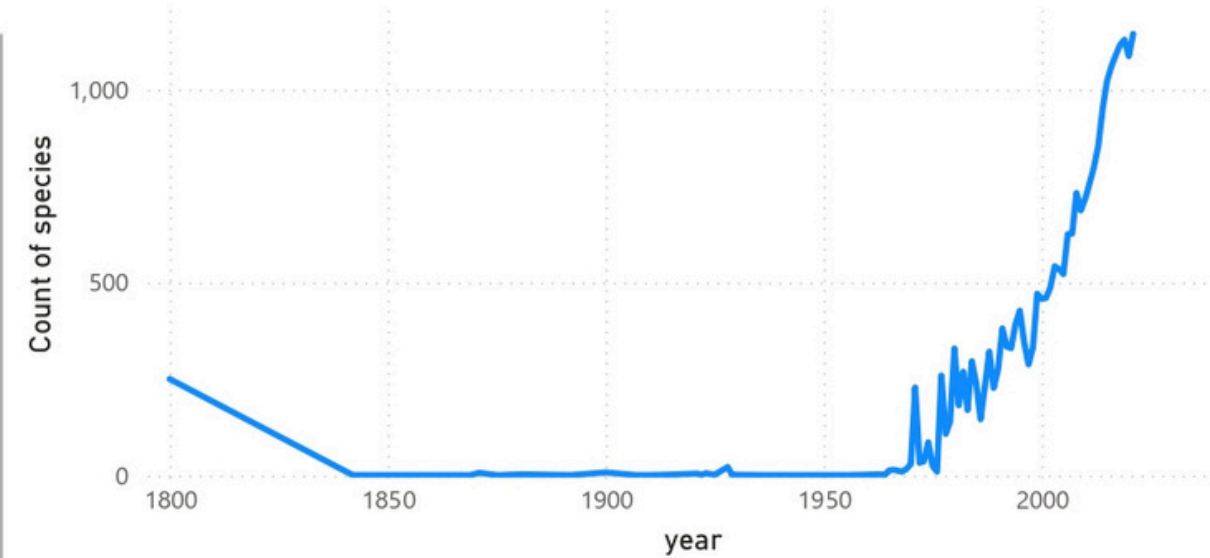
Wild Life Analytics - A Report On Bird Observations Across India

state, species

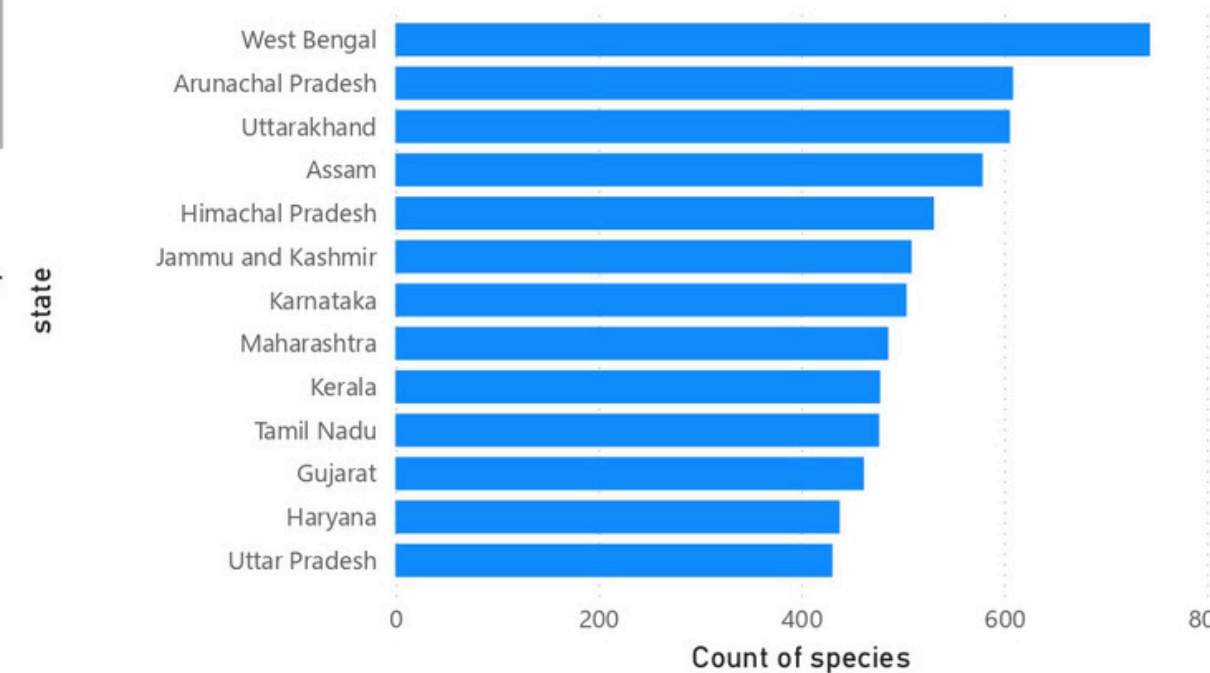
Search

- ☐ Andaman and Nic...
- ☐ Andhra Pradesh
- ☐ Arunachal Pradesh
- ☐ Assam
- ☐ Bihar
- ☐ Chandigarh
- ☐ Chhattisgarh
- ☐ Dadra and Nagar ...
- ☐ Daman and Diu
- ☐ Delhi
- ☐ Goa
- ☐ Gujarat
- ☐ Haryana
- ☐ Himachal Pradesh
- ☐ Jammu and Kashmir
- ☐ Jharkhand
- ☐ Karnataka
- ☐ Kerala
- ☐ Lakshadweep
- ☐ Madhya Pradesh
- ☐ Maharashtra

Count of species by year



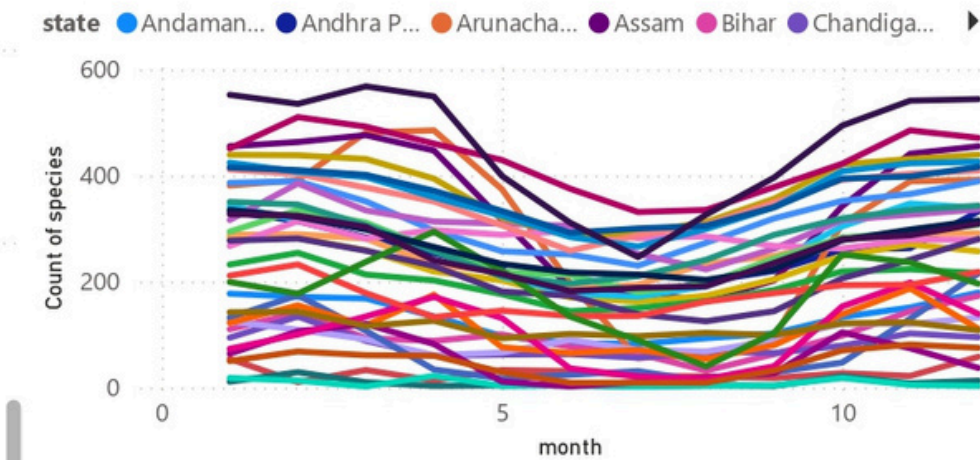
Count of species by state



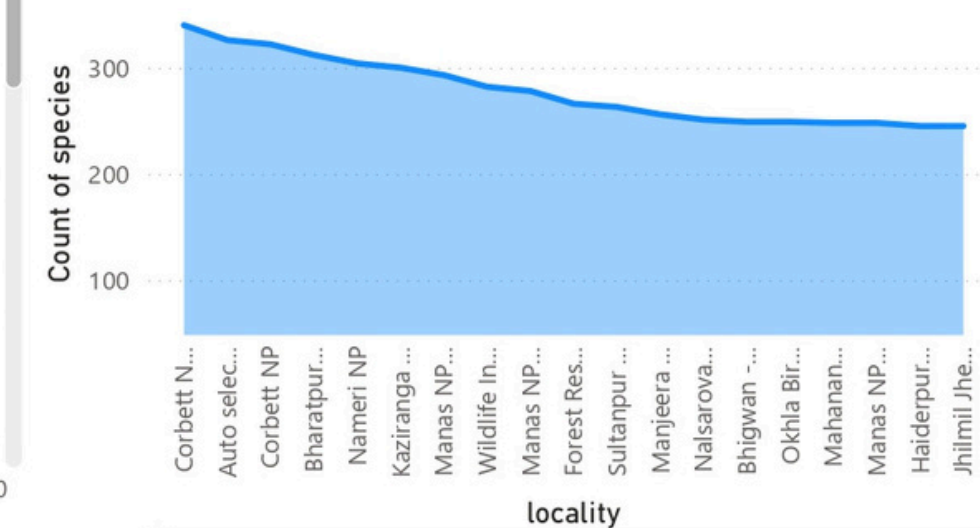
1247

Count of species

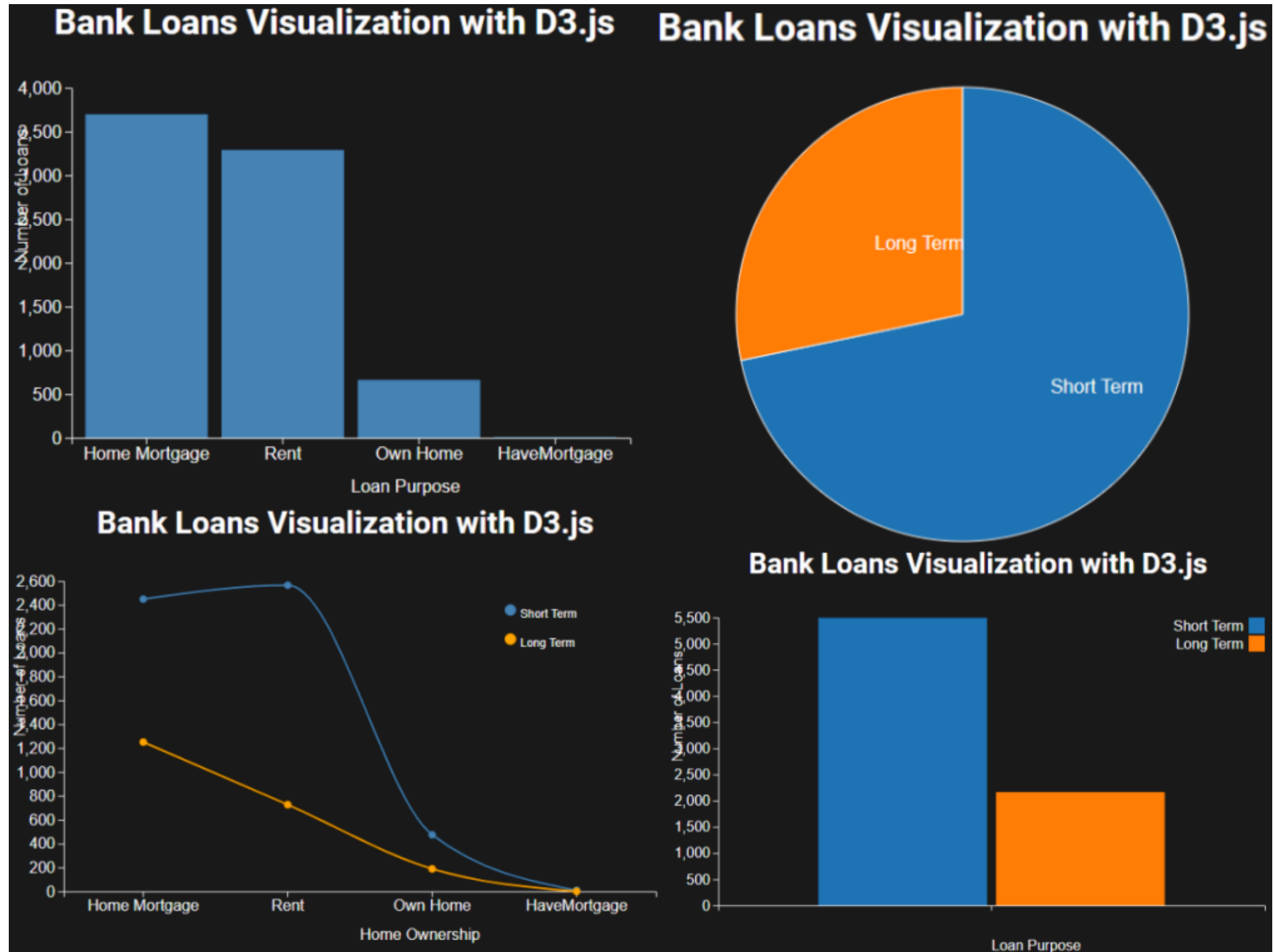
Count of species by month and state



Count of species by locality



EXPERIMENT 7: D3.JS



EXPERIMENT 8: D3.JS

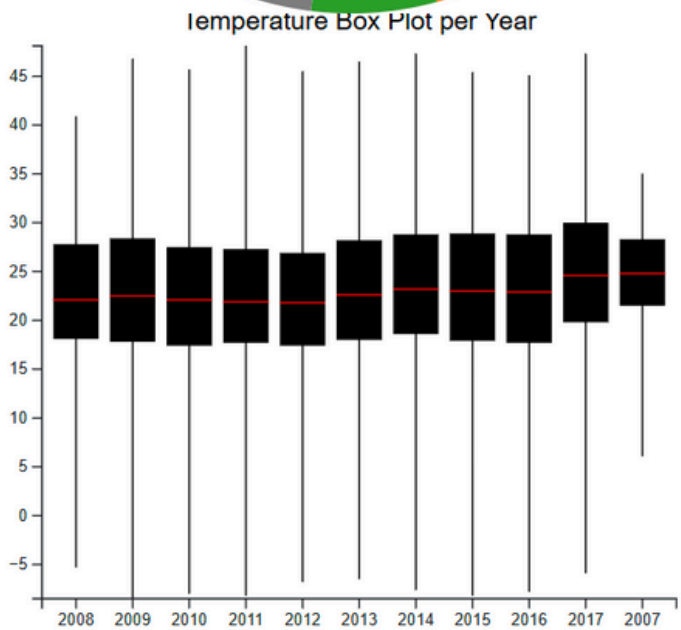
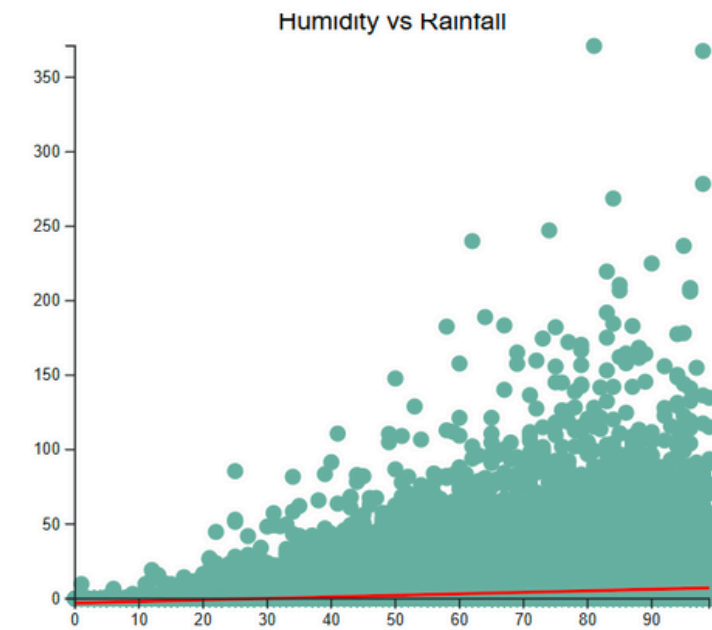
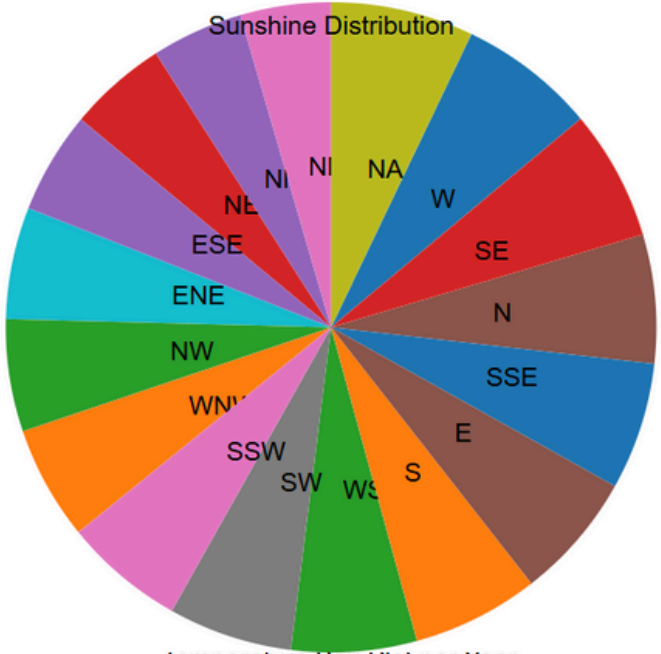
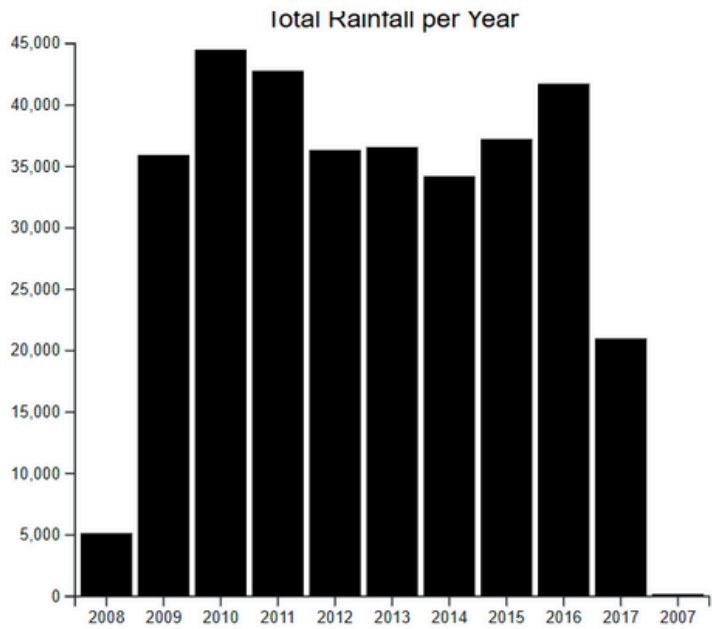
Weather Dashboard

Max Temperature:
48.1

Min Temperature:
-8.5

Max Humidity:
100

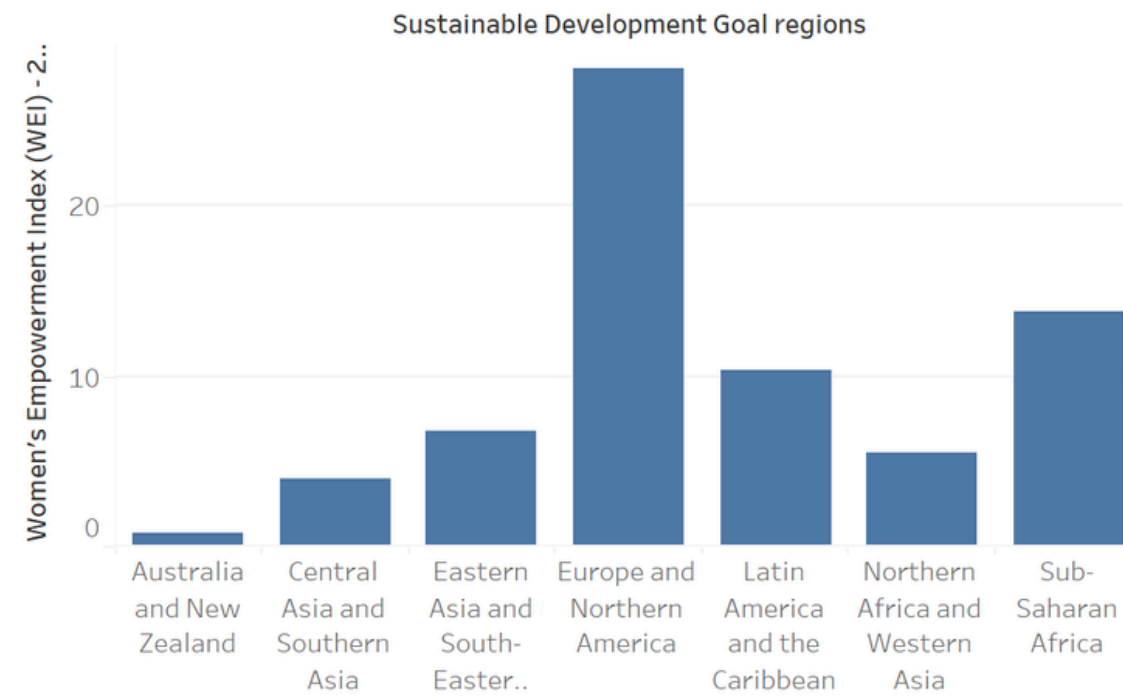
Location with Most
Rainfall:
Cairns (17157.20 mm)



EXPERIMENT 9: TABLEAU

Women Empowerment Analytics

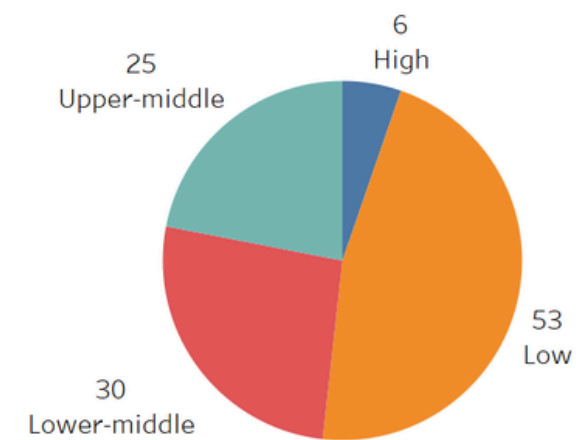
Regional Average Women's Empowerment Index



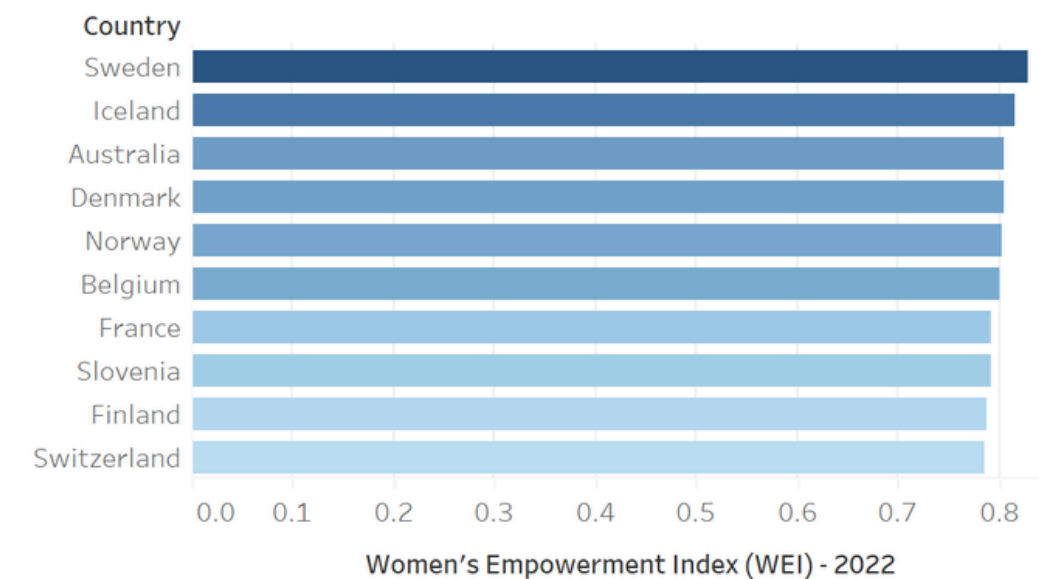
WEI vs Global Gender Parity Index Correlation



Distribution by Women's Empowerment Groups



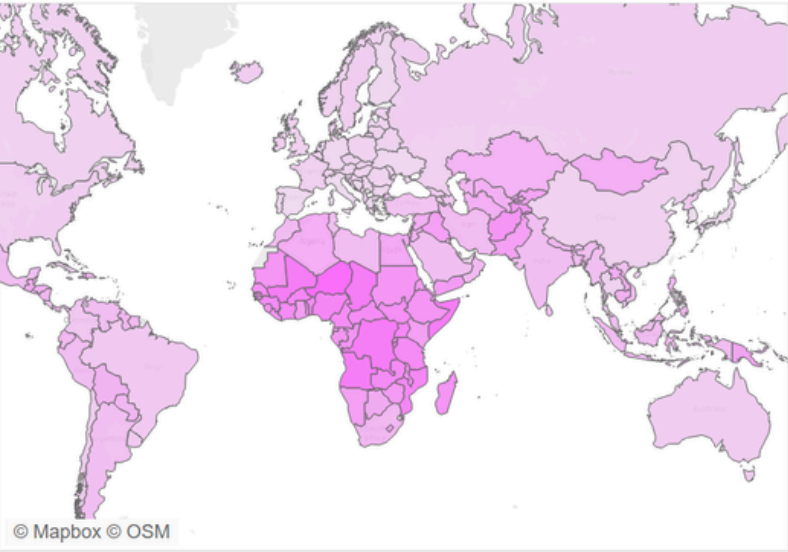
Top 10 Countries by Women's Empowerment Index



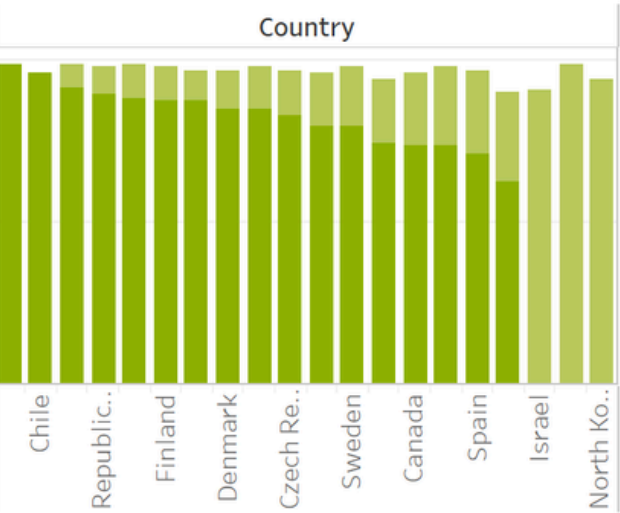
EXPERIMENT 10: TABLEAU

Global Education Statistics

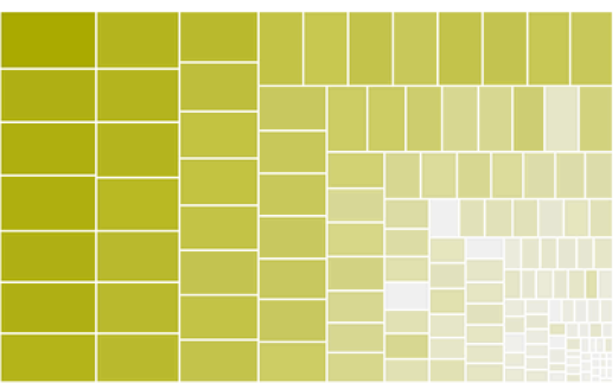
Countrywise Birth Rates



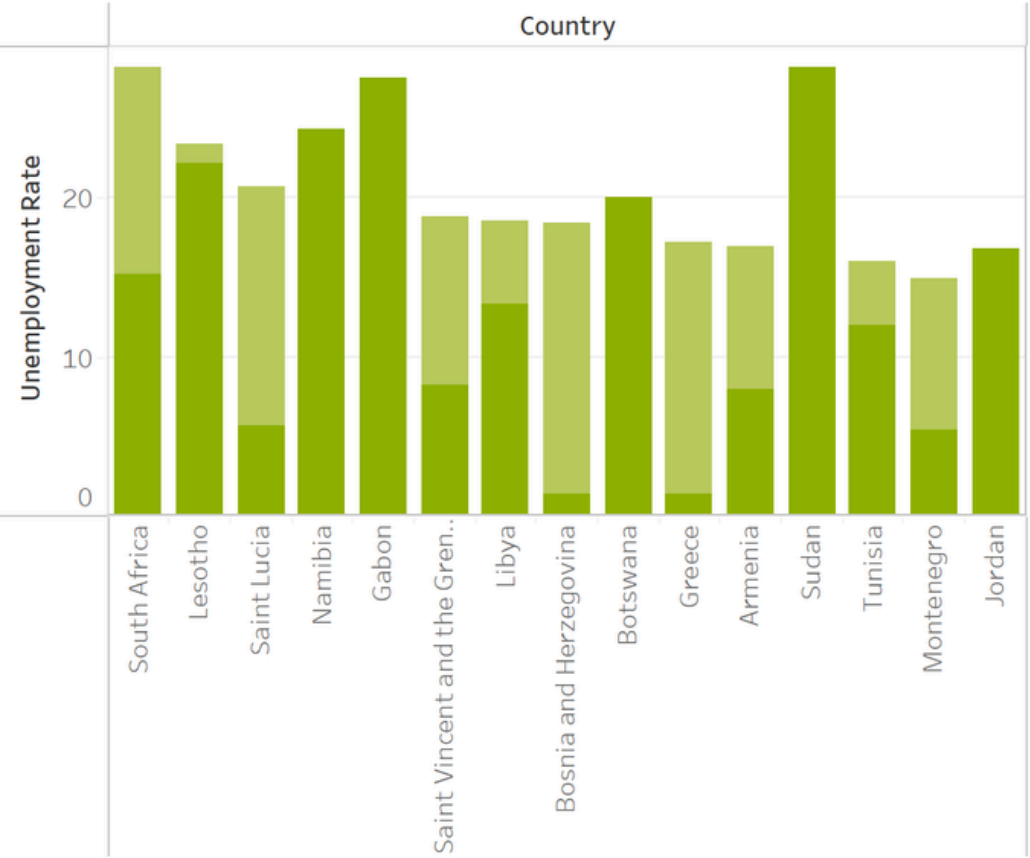
Countries with Highest Proficiency in Math & Reading



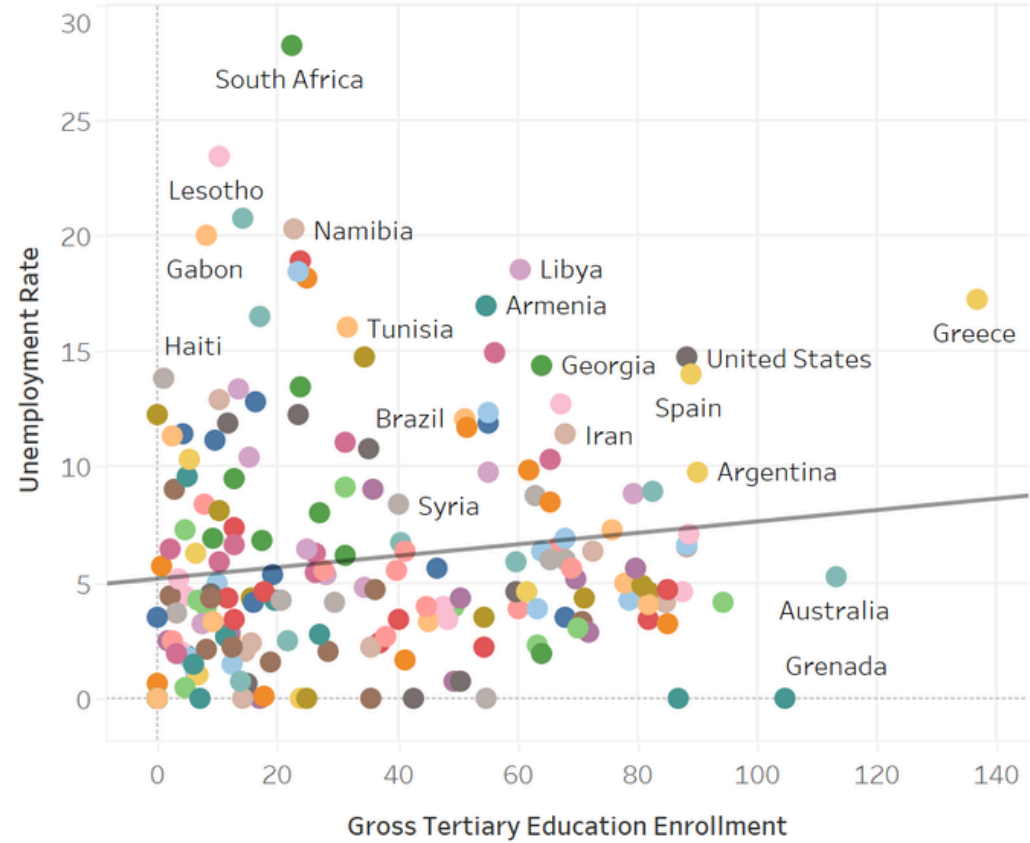
Pre-Primary Ages Males vs Females



Top 15 Countries By Unemployment Rate



Countries with Highest Educated Unemployed People



CONCLUSION

Throughout this series of experiments, I explored various data visualization tools and techniques, including Python, R, Power BI, Tableau, and D3.js. Each tool provided unique strengths in creating both basic and advanced visualizations. Power BI & Tableau offer interactive dashboards and storytelling. Python and R enabled in-depth analysis with precise control over complex charts, while D3.js provided the flexibility to create fully customized, web-based visualizations. This comprehensive approach deepened my understanding of effective data visualization.

The background features a minimalist design with teal-colored geometric elements. In the top-left and bottom-left corners, there are nested rectangular outlines. In the top-right and bottom-right corners, there are clusters of small teal circles arranged in a grid-like pattern. A diagonal teal line runs from the top-left towards the bottom-right, intersecting the other elements.

THANK YOU

By Vineet Parmar